

HUMANOID ROBOTS

Physical AI is enabled by mechanical hardware

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We are entering an era where AI enables multipurpose applications, but the mechanical hardware must be done right.

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AI WILL HAVE ITS BIGGEST INFLUENCE THROUGH HUMAN-CENTRIC ROBOTS

As humanoids enter the public sphere, forecasters are trying to predict where the technology will go. Some are suggesting that hardware will become a commodity, leading to largely software-defined general-purpose machines, but this goes too far. Artificial intelligence (AI) is just software trapped behind the physics of the hardware it is running on. Existing industrial robotic automation solutions are better than humans at achieving high-precision position trajectories and, thus, better at machining metal, spot welding, or high-speed pick-and-place operations. However, humans remain much better at going over any sort of terrain, mobile manipulation, assembly tasks, and other things that require a soft, force-based approach to physical interaction. AI alone, or any software, will not bridge the gap between these two sets of capabilities.

HARDWARE DYNAMICS DRIVE WHAT IS PHYSICALLY POSSIBLE

It is easy to make a robot that looks like a human but much more difficult to make a robot that moves like a human, especially when making contact with the physical world. Although we can observe the morphology and motion trajectory of a person or robot, we do not have a complete understanding of how humans achieve such capable manipulation and locomotion and, thus, cannot yet recreate it. Even a robot that looks convincingly like a person may have very different dynamics—joint friction, inertia, or compliance—and those hardware dynamics are the only path through which any control approach, AI or otherwise, can act on the world. They are at least as important as the morphology of the robot and can be either a barrier to control or an essential partner, depending on their properties.

A human motion is more than a series of positions; it is also an entire set of dynamics that will respond in particular ways when encountering the environment. It is possible to generate trajectories for the robot to follow that look extremely convincing until some disturbance, like an unexpected step down, shows major performance and visual differences between the motions of the robot and a person. How can we make robot arms that swing freely for balance during locomotion, whose motion is determined by the pendular swing of their mass and near-passive joints, while still being able to apply large forces to lift heavy things? How can we place a foot gently while walking on uneven terrain, repeatedly making contact that is never perfectly anticipated? And how can we make fingers with the compliance, strength, and force sensitivity for dexterous manipulation?

PREDICTIONS FOR THE BEST HARDWARE PATH

Human-centric robots will be bipedal, with an upright torso, and bimanual. They will have human-robot interaction features like a face, but that is where similarities will end: There is no reason for human-centric machines to look just like us. In the future, looking back on some of today's humanoid robots with precise human dimensions or five-fingered hands will be akin to looking back at early heavier-than-air flying machine prototypes. Some had flapping wings and other birdlike biomimetic features, because we did not yet understand the physics of flying, just as the physics of locomanipulation are presently in their early days. However, once the industry well understands the physics of locomanipulation and the softer science of human-robot interaction, we will see a proliferation of human-centric machines with familiar features but distinct

differences, just as we see a proliferation of aircraft that still have wings (but not feathers).

The basic humanoid form comes from first principles. Dynamic stability (balance) is important for operating in small spaces like hallways while reaching up high, and legs can be far more stable than wheels, because the feet can more quickly move to new locations to support the center of mass when there is a disturbance. An upright torso further enables a narrow footprint while allowing space for computation and batteries and improving the ability to balance by leaning in the direction of acceleration. A pair of arms mounted at the highest points on the torso (the shoulder locations) enables a good workspace for lifting large objects from the ground to the highest possible shelf, a good workspace for catching a fall, and excellent inertial actuation for turning or balance.

Primary actuators will be electromagnetic, a refinement and evolution of the brushless linear dc motors that we see now, and will achieve much higher torque for a given motor mass than today's motors. We will see new motor forms that improve manufacturability along with torque density through new configurations like the YASA axial flux motors (*1*) used in cars now, transverse flux variants, or just refinements of a standard radial flux motor that are enabled by new three-dimensional electromagnetic modeling methods, new high-flux magnetically soft materials, and new manufacturing approaches. Even with higher torque density in motors, a transmission will always be necessary, especially for larger robots, because scaling laws mean that larger machines are proportionally weaker than smaller machines (ants can lift many times their weight, but elephants cannot jump). New classes and types of actuators, such as polymer muscles, are still far away, and pneumatic or hydraulic actuators have fundamental flaws in energy efficiency and performance in a humanoid application.

The transmission of choice for large joints (knee, elbow, etc.) will be rolling-contact cycloids. They offer an unmatched combination

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Agility's Digit robot, the first humanoid robot deployed commercially to do physical work.

of low friction, high durability, wear characteristics even with regular impacts, compact form factor, torque density for their mass, and manufacturability without exotic methods. They can be integrated well with electromagnetic actuators into a convenient module to be used at robot joints. In combination with the high torque density of new motors, nothing will match their high-bandwidth torque control and backdrivability.

Hands and fingers will be tendon driven, using series elastic actuators. Tendons, combined with links that are only in compression loading in a tensegrity-style approach, are the most structurally efficient way to apply large forces at the ends of thin digits (2). Actuators will be in the forearm, as the muscles that power human hands are, because of the high power and force needed. Given the extreme compactness and forces needed from hand actuators, even when moved to the forearm, speed reductions will be high, and they will not be directly backdrivable, as the large-joint cycloids can be. However, we will need exceptional force control and impact tolerance in the fingers; thus, series elasticity will be required (3). Last, tendons are not durable. They will be replaceable wear components, like the soles of the feet.

Walking will require a functional heel strike to absorb the impact of foot contact and transition from leg swing to a firm stance (4). To power the gait for efficient, faster walking, a toe-off will also be required. This is because a humanoid will have a near-straight knee most of the time so that it will not be using excessive energy to support its weight. The heel strike insulates the large leg motors from the initial

impact, applies torque about the ankle, and allows the knee to begin bending and absorb the rest of the impact, and the toe-off extends the leg beyond the straight-knee configuration to add energy with each stride. There will be structure in the foot to handle this well, beyond the flat metal plates that we see today.

LOOKING FORWARD

Humanoids are beginning to provide utility today in commercial and industrial environments, doing work that has been designed for humans, in spaces designed for humans. These first robot tasks are still process automated: They are more variable than typical robot tasks, and, thus, humans still do them, but they still meet the classic three “D’s” of robotics, being dull, dirty, or dangerous, like moving totes around, sorting packages, or simple kitting processes. As humanoids gain increasing capability and generality, become safe, and benefit from an increasingly detailed regulatory framework and as they come down in price, they will expand into many more commercial and industrial settings, including familiar retail environments. They will move gradually from the back room to more public spaces as certifiably safe operation is addressed. Safety, of course, relies not only on perception and context-aware intelligent choices but also on actuators that can reliably (a safety-certified, redundant level of reliably) apply safe forces in safe locations when they physically interact with people, by avoiding pinch points, falling on someone’s foot, and other physical risks (5, 6). Only after robot locomotion and manipulation have achieved

broadly general capabilities in environments as varied as people’s homes, have been scaled to the point where the cost is affordable to families, and have gradually improved and proven their safe operation to the point where we can trust them around our children and pets will they begin to enter homes and become part of everyday life for most people.

Humanoids, enabled by AI, bring a new era of robotics: unlocking operations wherever people work without requiring architectural modifications. Additionally, multipurpose robots will address a variety of tasks that could not be automated easily with single-purpose machinery. These two features, operating in human spaces and being multipurpose, are going to redefine labor and amplify human ambition in a way that will be analogous to an accelerated industrial revolution. There will be an array of humanoid robots as varied as vehicles on the road, with their own specialties and features. Humans will soon have robot helpers in a way that humanity has never experienced before.

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